

Centre No.						Paper Reference						Surname	Initial(s)	
Candidate No.						6	6	7	7	/	0	1	Signature	

Paper Reference(s)

6677/01

Edexcel GCE

Mechanics M1

Advanced/Advanced Subsidiary

Friday 6 June 2014 – Afternoon

Time: 1 hour 30 minutes

Examiner's use only

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Team Leader's use only

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[illegible]

Materials required for examination

Mathematical Formulae (Pink)

Items included with question papers

Nil

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation or symbolic differentiation/integration, or have retrievable mathematical formulae stored in them.

Instructions to Candidates

In the boxes above, write your centre number, candidate number, your surname, initials and signature. Check that you have the correct question paper.

Answer ALL the questions.

You must write your answer to each question in the space following the question.

Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either two significant figures or three significant figures.

When a calculator is used, the answer should be given to an appropriate degree of accuracy.

Information for Candidates

A booklet 'Mathematical Formulae and Statistical Tables' is provided.

Full marks may be obtained for answers to ALL questions.

The marks for individual questions and the parts of questions are shown in round brackets: e.g. (2).

There are 7 questions in this question paper. The total mark for this paper is 75.

There are 28 pages in this question paper. Any blank pages are indicated.

Advice to Candidates

You must ensure that your answers to parts of questions are clearly labelled.

You should show sufficient working to make your methods clear to the Examiner.

Answers without working may not gain full credit.

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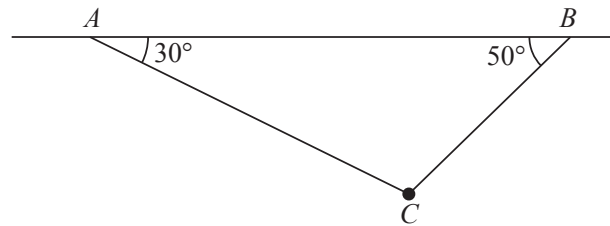


Figure 1

A particle of weight W newtons is attached at C to two light inextensible strings AC and BC . The other ends of the strings are attached to fixed points A and B on a horizontal ceiling. The particle hangs in equilibrium with AC and BC inclined to the horizontal at 30° and 50° respectively, as shown in Figure 1.

Given that the tension in BC is 6 N, find

- (a) the tension in AC , **(3)**
- (b) the value of W . **(3)**

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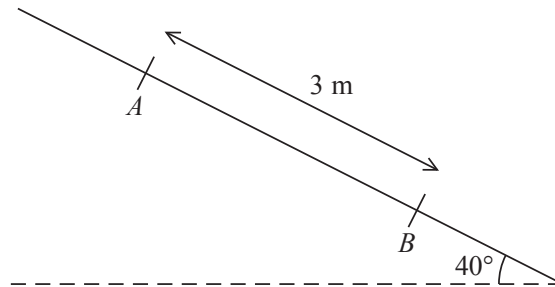


Figure 2

A rough plane is inclined at 40° to the horizontal. Two points A and B are 3 metres apart and lie on a line of greatest slope of the inclined plane, with A above B , as shown in Figure 2. A particle P of mass m kg is held at rest on the plane at A . The coefficient of friction between P and the plane is $\frac{1}{2}$. The particle is released.

- (a) Find the acceleration of P down the plane.

(5)

- (b) Find the speed of P at B .

(2)



Question 2 continued



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3. A ball of mass 0.3 kg is released from rest at a point which is 2 m above horizontal ground. The ball moves freely under gravity. After striking the ground, the ball rebounds vertically and rises to a maximum height of 1.5 m above the ground, before falling to the ground again. The ball is modelled as a particle.
- (a) Find the speed of the ball at the instant before it strikes the ground for the first time. (2)
- (b) Find the speed of the ball at the instant after it rebounds from the ground for the first time. (2)
- (c) Find the magnitude of the impulse on the ball in the first impact with the ground. (2)
- (d) Sketch, in the space provided, a velocity-time graph for the motion of the ball from the instant when it is released until the instant when it strikes the ground for the second time. (3)
- (e) Find the time between the instant when the ball is released and the instant when it strikes the ground for the second time. (4)



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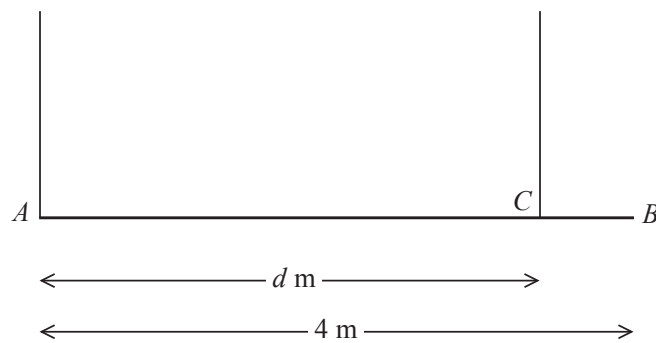


Figure 3

A beam AB has weight W newtons and length 4 m. The beam is held in equilibrium in a horizontal position by two vertical ropes attached to the beam. One rope is attached to A and the other rope is attached to the point C on the beam, where $AC = d$ metres, as shown in Figure 3. The beam is modelled as a uniform rod and the ropes as light inextensible strings. The tension in the rope attached at C is double the tension in the rope attached at A .

- (a) Find the value of d .

(6)

A small load of weight kW newtons is attached to the beam at B . The beam remains in equilibrium in a horizontal position. The load is modelled as a particle. The tension in the rope attached at C is now four times the tension in the rope attached at A .

- (b) Find the value of k .

(6)

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Question 4 continued



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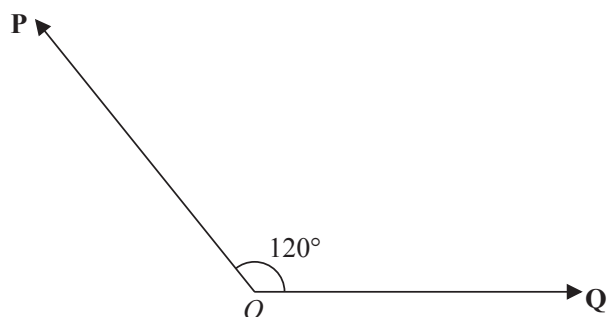


Figure 4

Two forces **P** and **Q** act on a particle at *O*. The angle between the lines of action of **P** and **Q** is 120° as shown in Figure 4. The force **P** has magnitude 20 N and the force **Q** has magnitude *X* newtons. The resultant of **P** and **Q** is the force **R**.

Given that the magnitude of $\mathbf{R}$ is $3X$ newtons, find, giving your answers to 3 significant figures

- (a) the value of X ,
- (5)**

- (b) the magnitude of $(\mathbf{P} - \mathbf{Q})$.

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Question 6 continued



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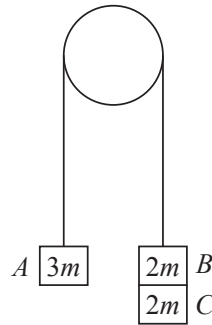


Figure 5

Three particles A , B and C have masses $3m$, $2m$ and $2m$ respectively. Particle C is attached to particle B . Particles A and B are connected by a light inextensible string which passes over a smooth light fixed pulley. The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5. The system is released from rest and A moves upwards.

- (a) (i) Show that the acceleration of A is $\frac{g}{7}$
- (ii) Find the tension in the string as A ascends.
- (7)
- At the instant when A is 0.7 m above its original position, C separates from B and falls away. In the subsequent motion, A does not reach the pulley.
- (b) Find the speed of A at the instant when it is 0.7 m above its original position.
- (2)
- (c) Find the acceleration of A at the instant after C separates from B .
- (4)
- (d) Find the greatest height reached by A above its original position.
- (3)



Question 7 continued

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Handwriting practice area with horizontal lines.

Q7

(Total 16 marks)

TOTAL FOR PAPER: 75 MARKS

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